Digital Assignment Lab - 4

Course Name :- Agile Development Process and Devops Lab
Course code :- ISWE406P
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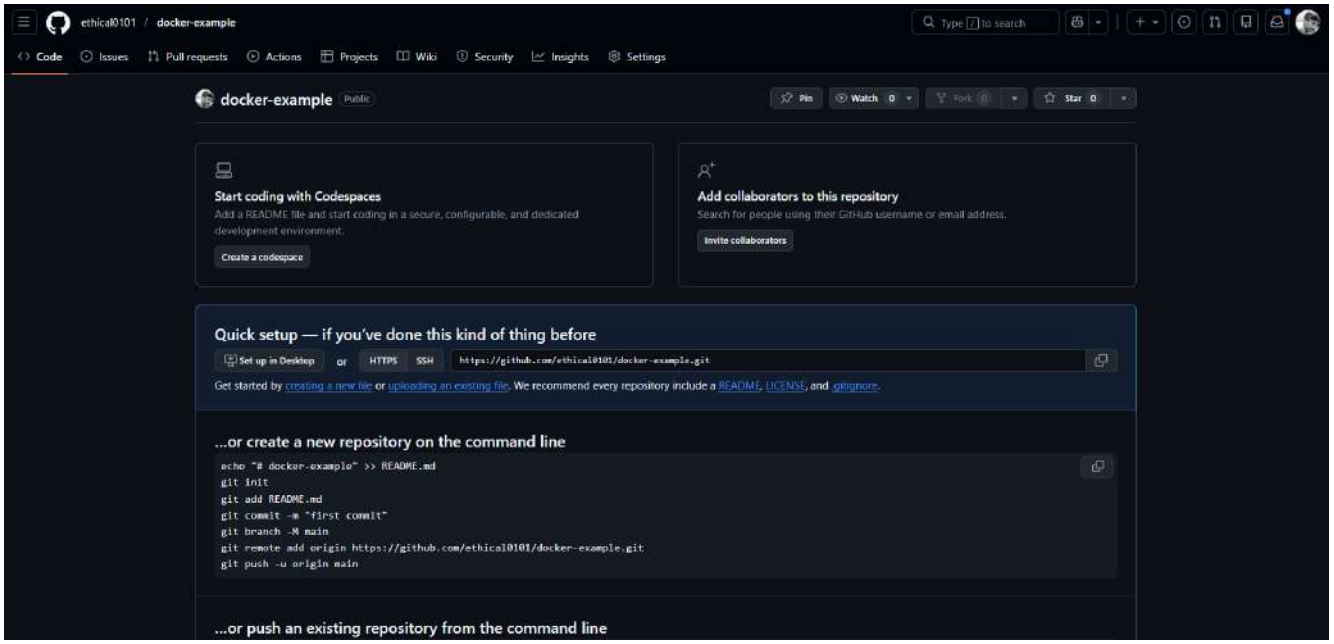
STEP 1:Create GitHub Repository

- **Steps:**
  - Go to GitHub
  - Click **New Repository**
  - Name → docker-example
  - Click **Create**



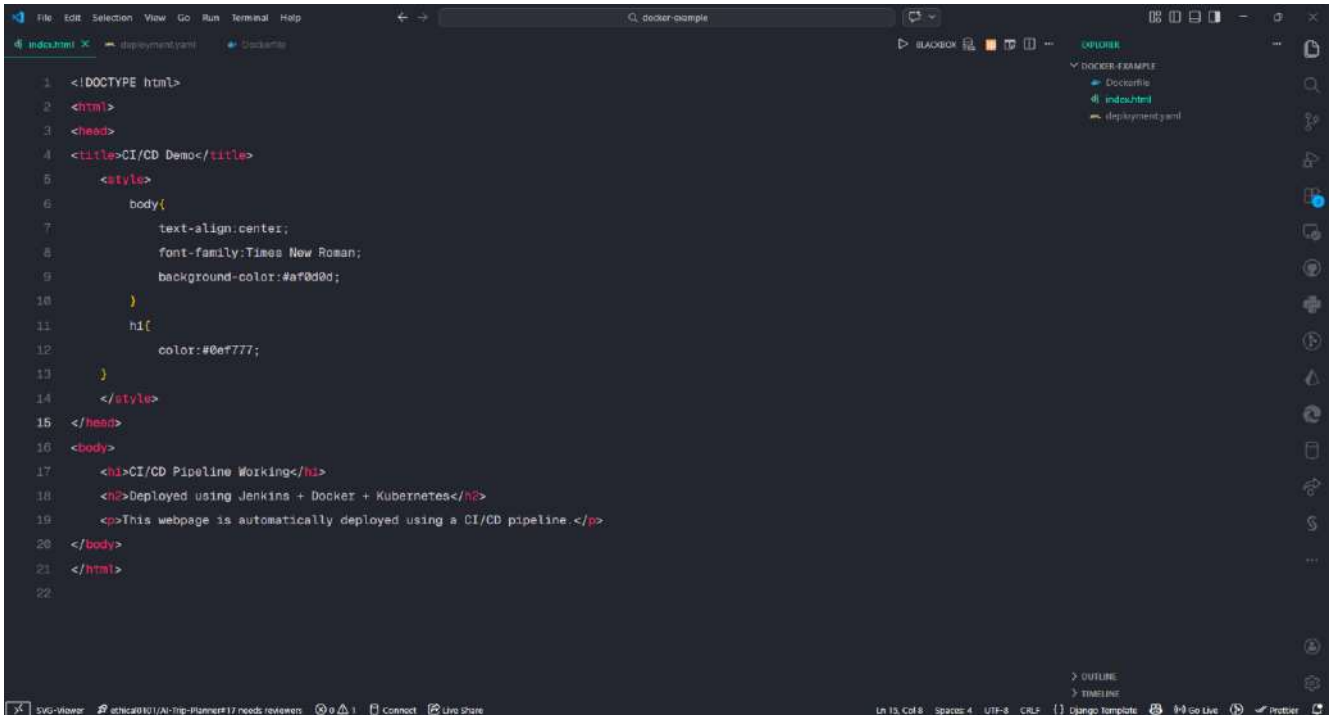
STEP 2:Create Files Locally

- **Create folder:**

docker-example
Inside create:

- index.html
- Dockerfile
- deployment.yaml

```
PS E:\> mkdir docker-example


    Directory: E:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        3/25/2026   9:56 PM                docker-example


PS E:\>
```



index.html

```
<!DOCTYPE html>
<html>
<head>
<title>CI/CD Demo</title>
   <style>
      body{
         text-align:center;
         font-family:Times New Roman;
         background-color:#af0d0d;
      }
      h1{
         color:#0ef777;
   }
   </style>
</head>
<body>
   <h1>CI/CD Pipeline Working</h1>
   <h2>Deployed using Jenkins + Docker + Kubernetes</h2>
   <p>This webpage is automatically deployed using a CI/CD pipeline.</p>
</body>
</html>
```

Dockerfile

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: html-demo
spec:
  replicas: 2
```

```
  selector:
    matchLabels:
      app: html-demo
  template:
    metadata:
      labels:
        app: html-demo
    spec:
      containers:
      - name: html-demo
        image: druthendra/html-demo:latest
        ports:
        - containerPort: 80

---
apiVersion: v1
kind: Service
metadata:
  name: html-demo-service
spec:
  type: NodePort
  selector:
    app: html-demo
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30010
```

STEP 3: Push Code to GitHub

```
PS E:\docker-example> git init
PS E:\docker-example> git add .
PS E:\docker-example> git commit -m "first commit"
[main (root-commit) 37a2d54] first commit
 3 files changed, 58 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 deployment.yaml
 create mode 100644 index.html
PS E:\docker-example> git branch -M main
PS E:\docker-example> git remote add origin https://github.com/ethical0101/docker-example.git
PS E:\docker-example> git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 870 bytes | 870.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ethical0101/docker-example.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS E:\docker-example>
```

STEP 4: Enable Kubernetes

• Open the Docker Dashboard: Launch the Docker Desktop application on your system.

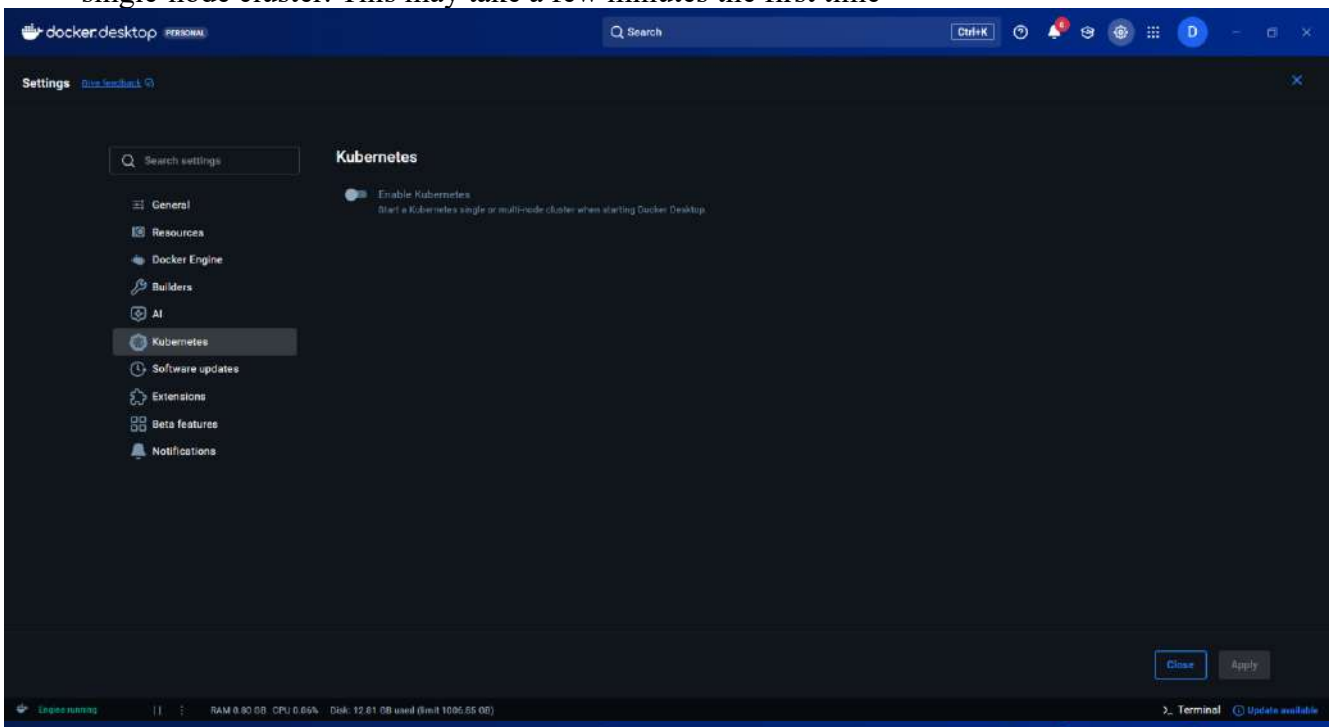
• Navigate to Settings: Click the gear icon (Settings or Preferences) in the top-right or sidebar of the Docker Dashboard.

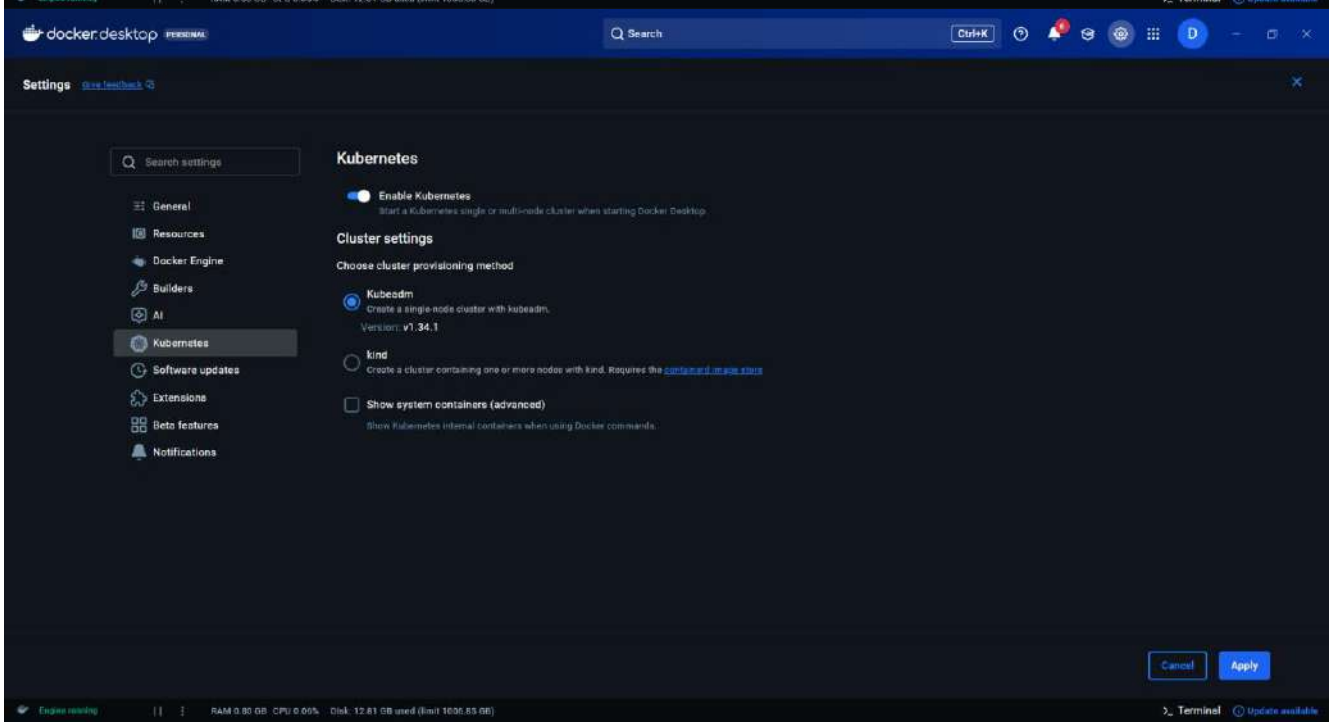
• Select the Kubernetes Tab: In the Settings menu, find and select the Kubernetes tab.

• Enable Kubernetes: Check the box labeled Enable Kubernetes.

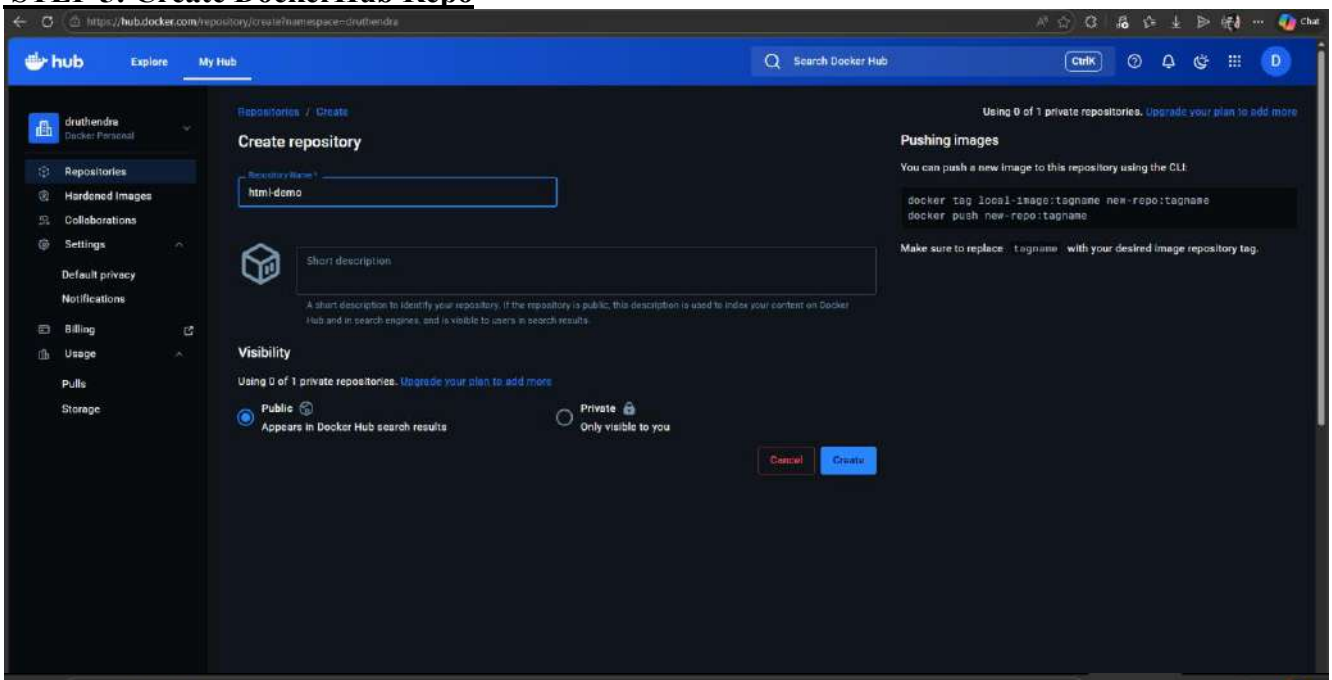
• Apply & Restart: Click the Apply & Restart button at the bottom of the settings window.

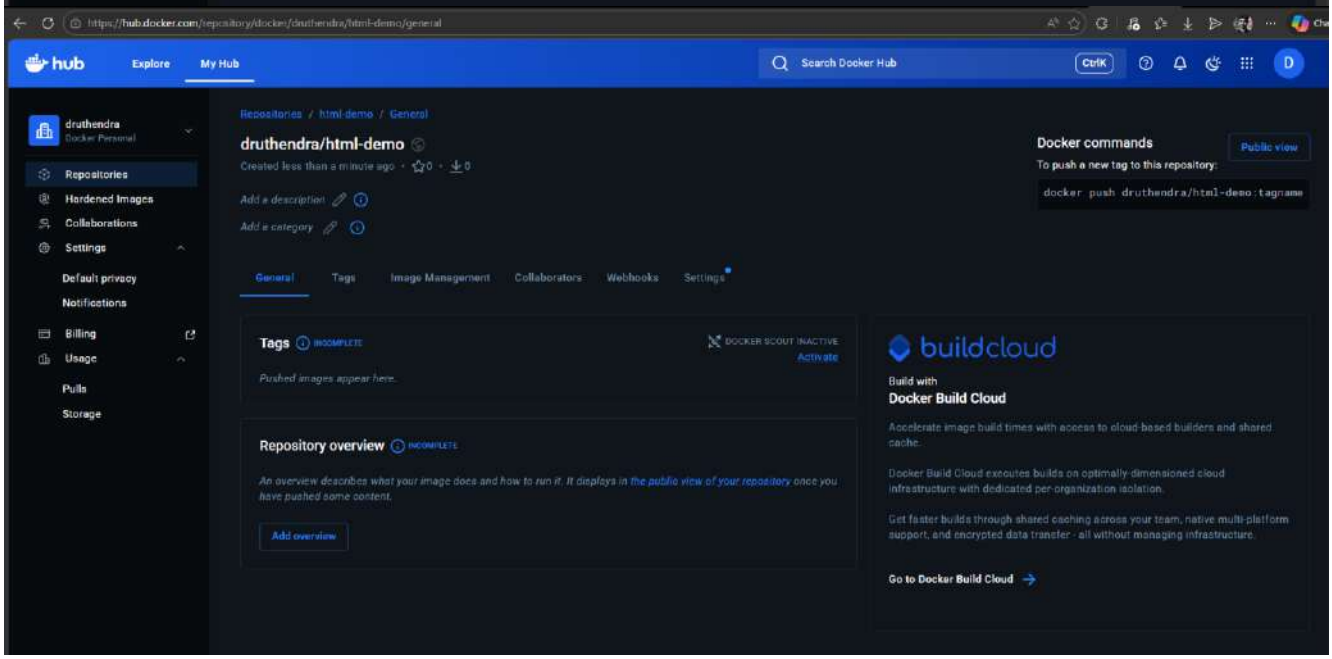
Docker Desktop will then download the necessary Kubernetes components and start a single-node cluster. This may take a few minutes the first time



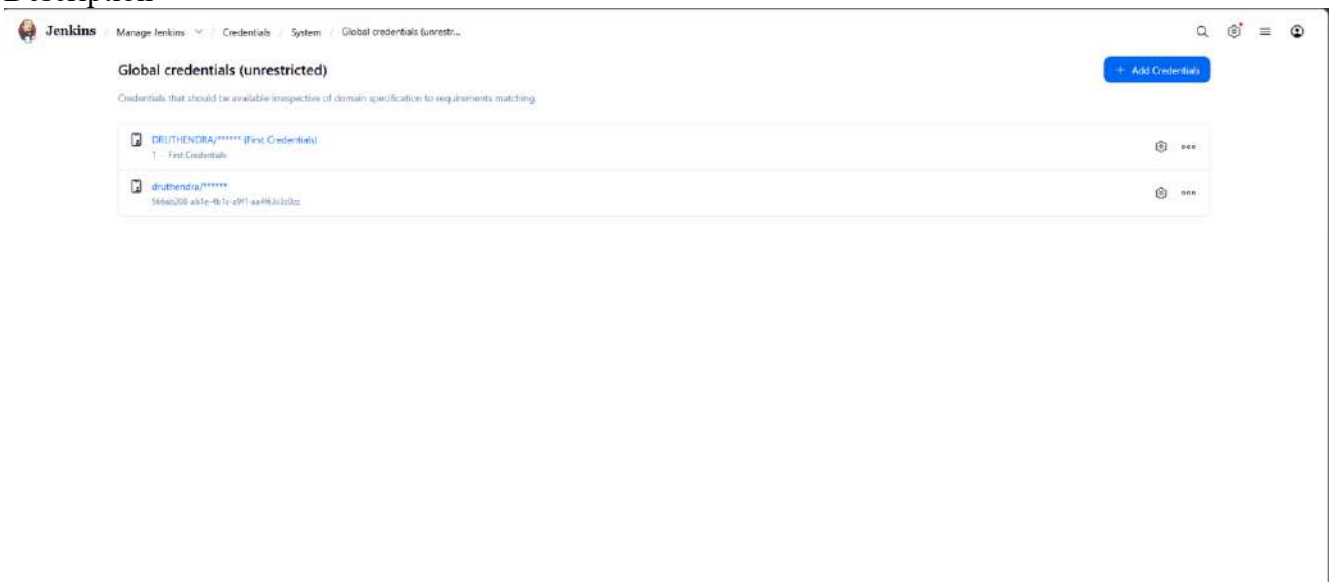


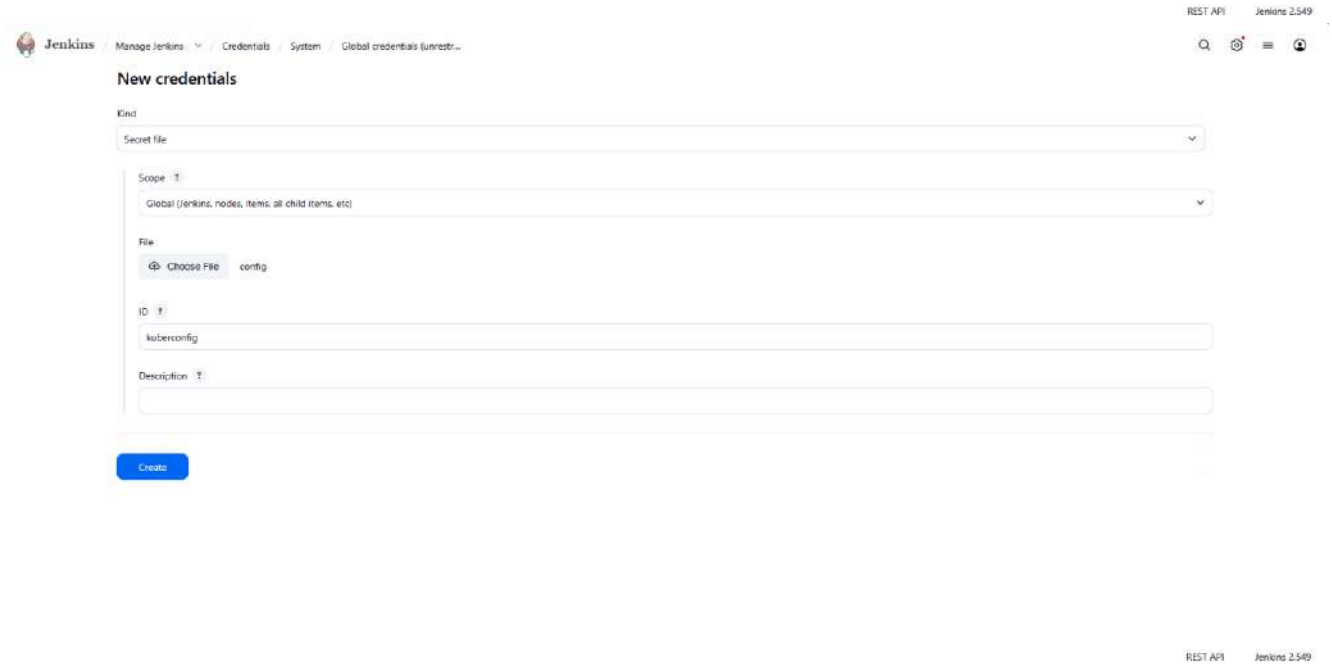
STEP 5: Create DockerHub Repo



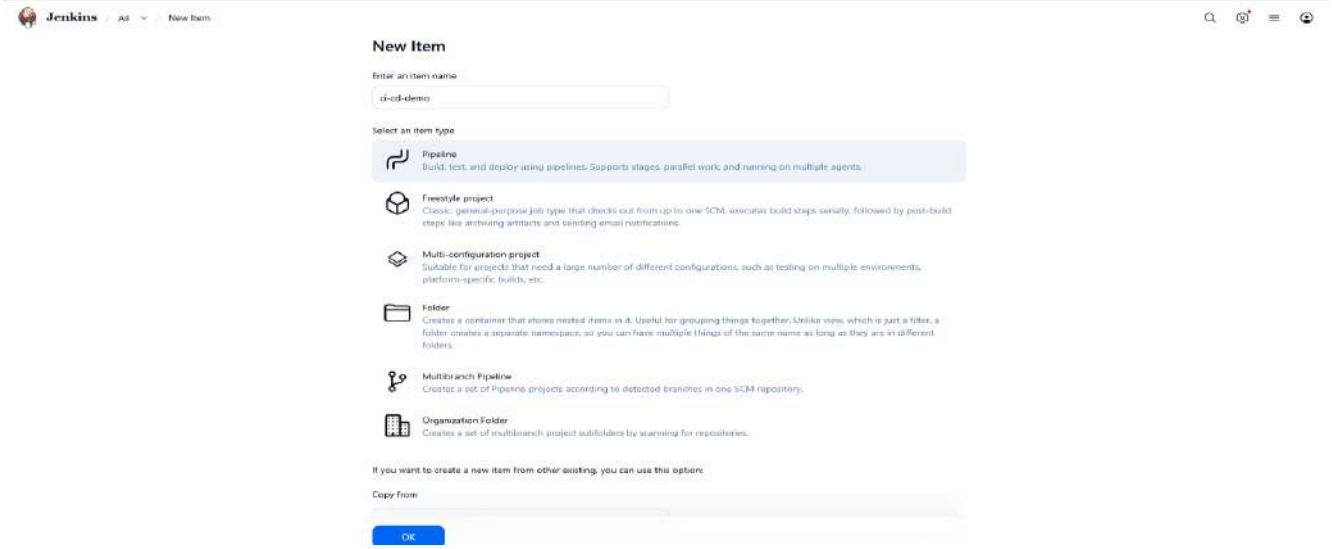


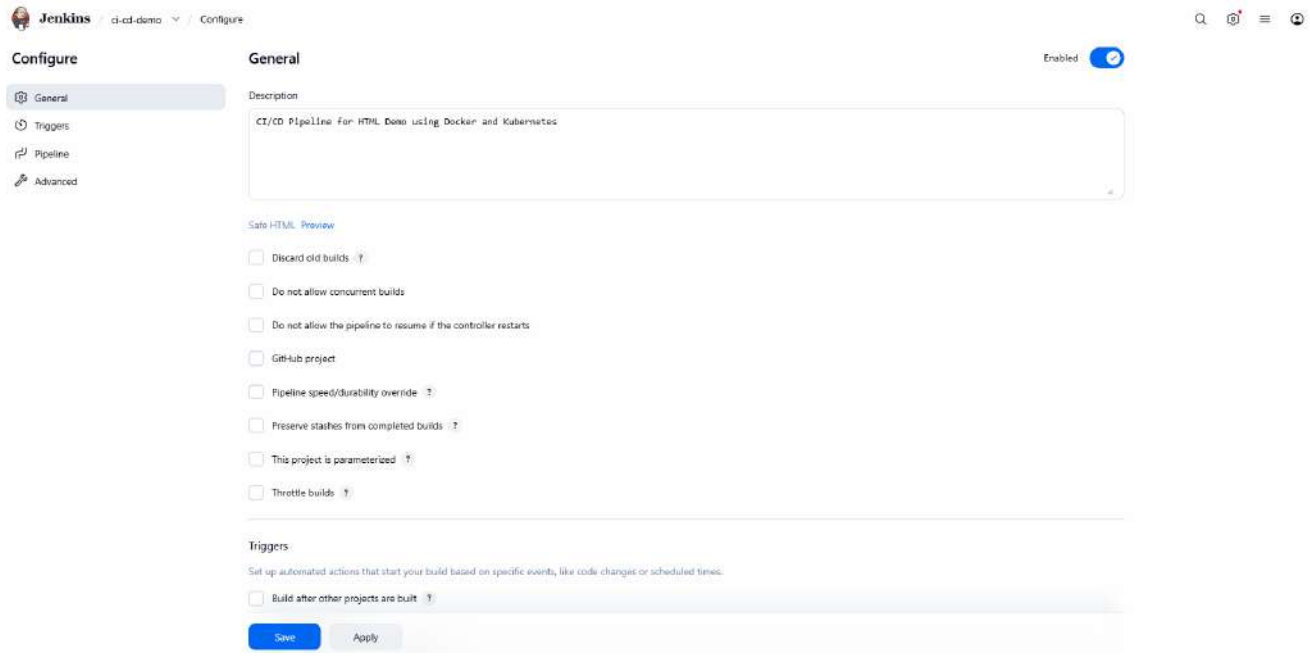
Open Jenkins In Jenkins Setting --> Credentials New Credentials – User name: , password , ID:, Description





Create New Item -> Pipeline Paste the code in pipeline script - - - - Update URL of GIT repository Make sure credentialsId: “dockerhub” same as mentioned in credentials Make sure that the ID mentioned in credentials “kuberconfig” is same as in script Run the Build Pipeline:

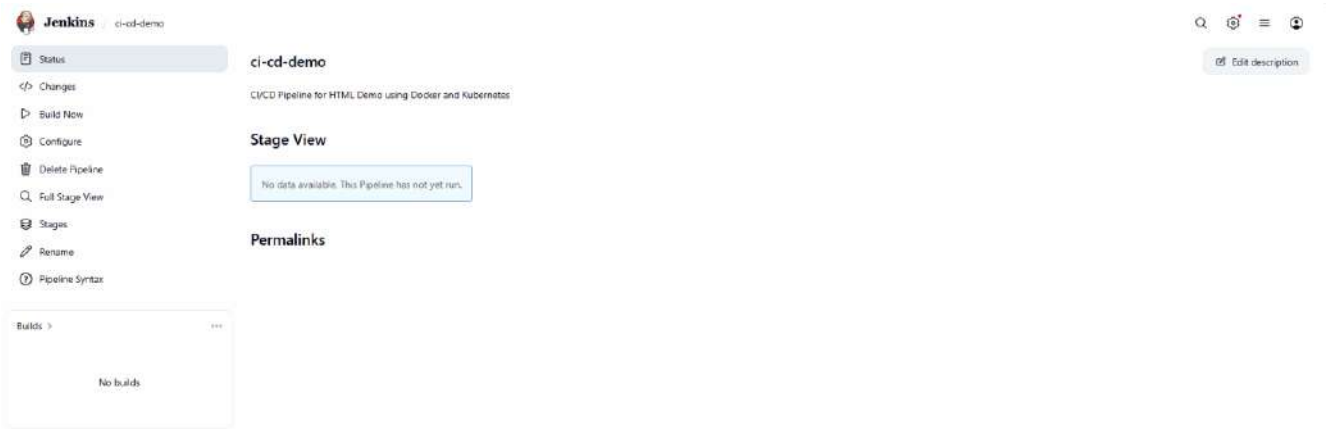




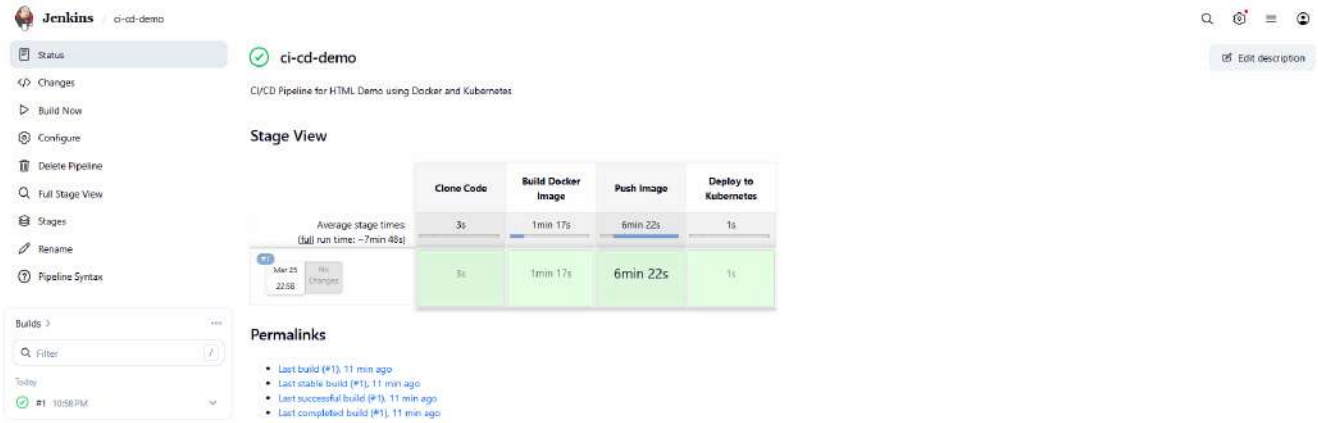
Pipeline Code:-

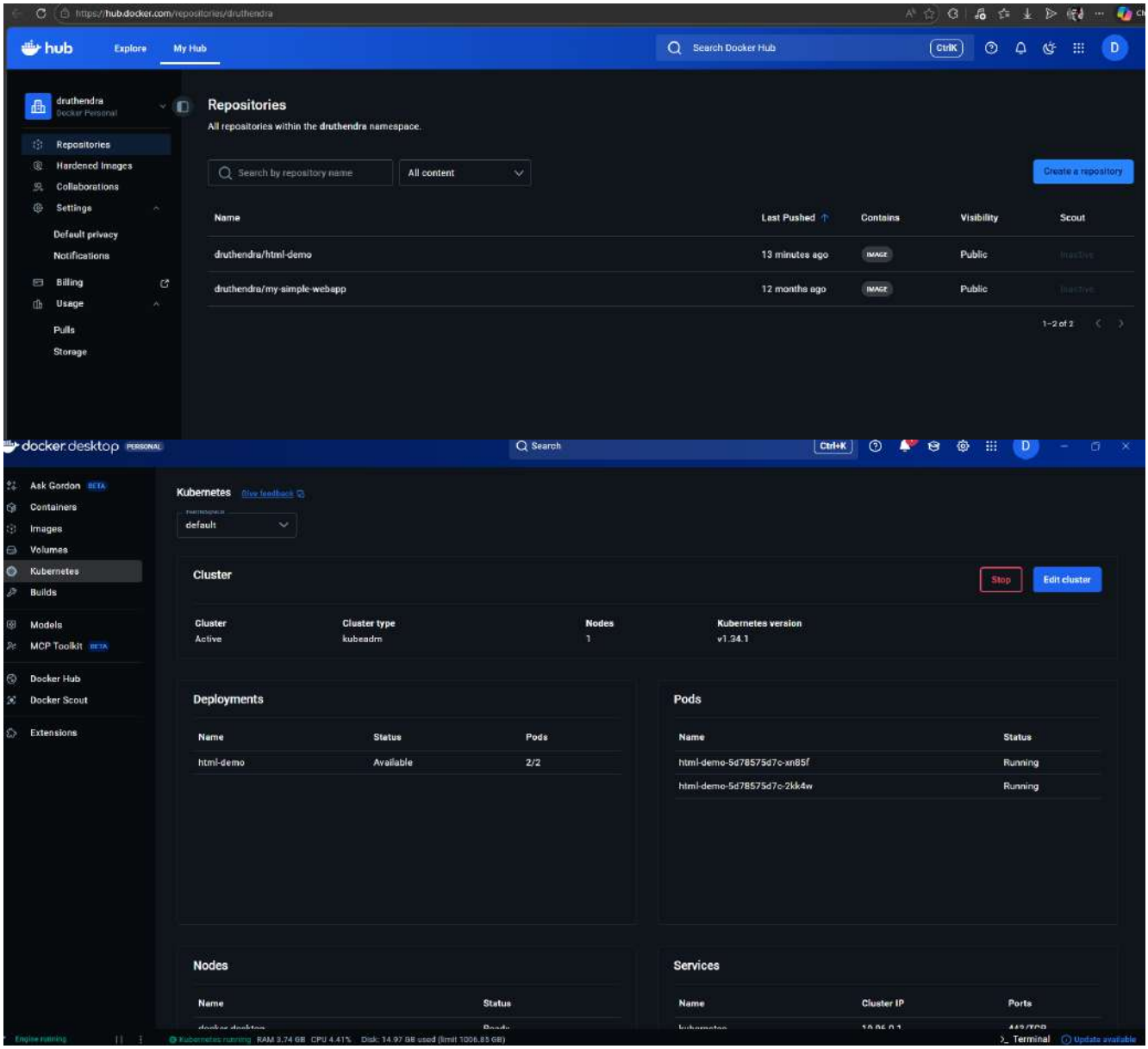
```
pipeline {
   agent any

   environment {
      DOCKER_IMAGE = "druthendra/html-demo"
   }

   stages {

      stage('Clone Code') {
         steps {
            git branch: 'main',
            url: 'https://github.com/ethical0101/docker-example.git'
         }
      }

      stage('Build Docker Image') {
         steps {
            bat 'docker build -t %DOCKER_IMAGE% .'
         }
      }

      stage('Push Image') {
         steps {
            withCredentials([usernamePassword(
               credentialsId: 'dockerhub',
               usernameVariable: 'USER',
               passwordVariable: 'PASS')]) {

               bat 'docker login -u %USER% -p %PASS%'
               bat 'docker push %DOCKER_IMAGE%'
            }
         }
      }

      stage('Deploy to Kubernetes') {
```

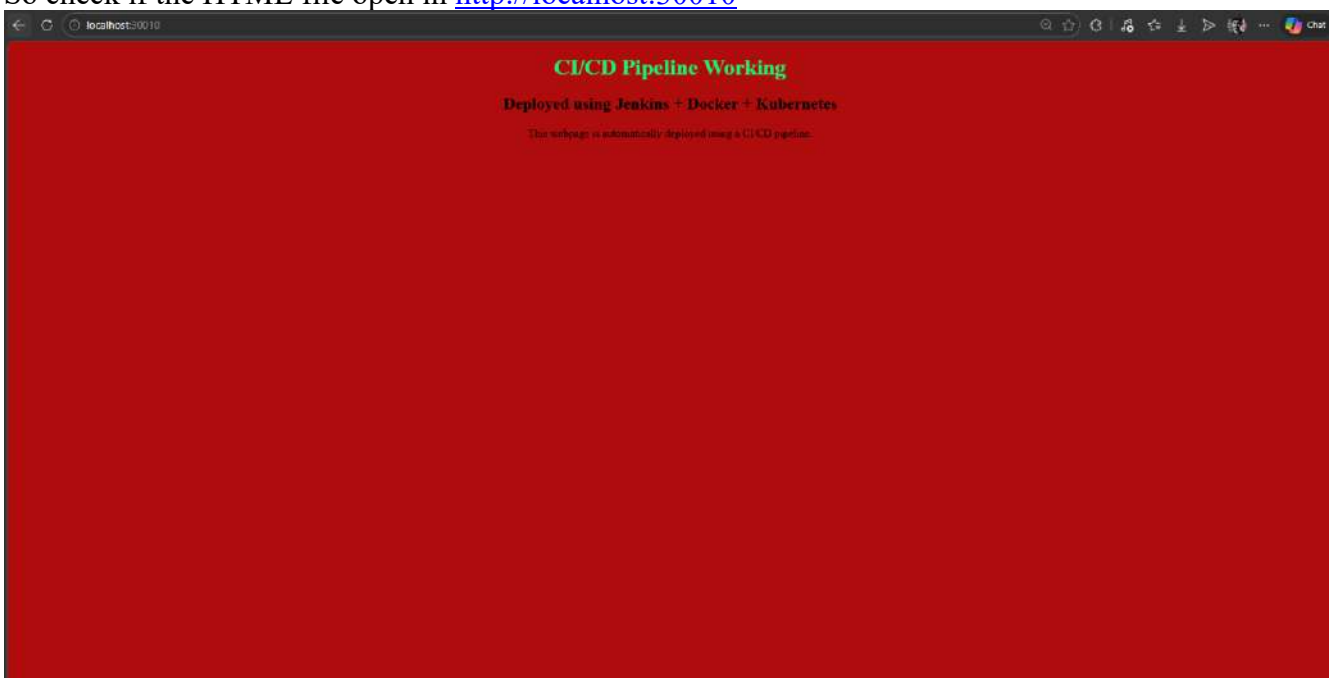
```
        steps {
            withCredentials([file(
                credentialsId: 'kuberconfig',
                variable: 'KUBECONFIG')]) {

                bat '''
                set KUBECONFIG=%KUBECONFIG%
                kubectl apply -f deployment.yaml --validate=false
                '''
            }
        }
    }
  }
}
```



Building:-





Also the HTML page should open in the port mentioned in deployment.yaml file: nodePort: 30010

So check if the HTML file open in http://localhost:30010